

1. (20 points) Short Answer. Each is worth 5 points.

- (a) List all elements of order 4 in $\mathbb{Z}_4 \oplus \mathbb{Z}_4$.
- (b) What does (123) get mapped to under the inner automorphism $\phi_{(234)}$ in S_4 ?
- (c) What is the order of $14 + \langle 8 \rangle$ in $\mathbb{Z}_{24} / \langle 8 \rangle$?
- (d) If $\phi : \mathbb{Z}_{10} \rightarrow \mathbb{Z}_{10}$ is an isomorphism with $\phi(1) = 7$, determine $\phi(5)$.

2. (10 points) Let $G = \mathbb{Z}_5 \oplus \mathbb{Z}_6$.

- Is G cyclic? Why or why not?
- If G is cyclic, list all of its generators. If G is not cyclic, what is the maximum order of an element in G ?

3. (14 points) Consider $H = \langle (12) \rangle$ in S_3 .

- List all left cosets of H .
- List all right cosets of H .
- Based on the cosets above, is H normal in G ? Explain.

4. (22 points) Consider $U(1400)$.

- Write $U(1400)$ as the external direct product of cyclic groups.
- What is the order of $U(1400)$?
- Does $U(1400)$ have an element of order 30? Explain.
- How many elements of order 2 are in $U(1400)$?
- How many subgroups of order 5 does $U(1400)$ have? Explain.

5. (3 points each) Determine whether the following is true or false. No explanation is required.

- (a) Every group of order n is isomorphic to some subgroup of the symmetric group S_n .
- (b) A subgroup H of a group G is a normal subgroup if and only if the number of left cosets of H is equal to the number of right cosets of H .
- (c) Every Abelian group of order 45 has a subgroup of order 5.
- (d) Some Abelian group of order 45 has a subgroup of order 10.
- (e) If $aH = Ha$ for all $a \in G$, then G is Abelian.
- (f) If G is a group of order 36, and H is a subgroup of G of order 18, then H is normal.
- (g) If G/H is a cyclic group, then G is a cyclic group.

6. (40 points) Answer the following questions. Provide an explanation of the answer using between 2 and 4 sentences. Each is worth 10 points. **Your answer will be graded on both the answer and the clarity of your solution.**

(a) What are possibilities for $\phi(1)$ in $\text{Aut}(\mathbb{Z}_{10})$?

(b) Each of $\mathbb{Z}_{12}$, D_6 , and A_4 have order twelve. Why are no two of these groups isomorphic?

(c) Let G be a group with subgroups H and K such that $|H| = 36$ and $|K| = 25$. What are the possible orders of $H \cap K$?

(d) Can a group of order 55 have 20 elements of order 11?

7. (20 points) Provide a clear proof that $\mathbb{R} \oplus \mathbb{R}$ is isomorphic to $\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}$.

8. (16 points) Prove the Normal Subgroup Test: Let H be a subgroup of G . Then H is normal in G if and only if for all $x \in G$, $xHx^{-1} \subseteq H$.

9. (12 points) Choose one of the following to prove. Mark the statement you are choosing to prove.

- If G be a group of order pq (where both p and q are primes), then either $Z(G) = e$ or that G is Abelian.
- If G be a group of order pq (where both p and q are primes), then G must have an element of order p and an element of order q .